



PROJECT DOCUMENTATION



OptiCrop – Project Documentation

 DATE
17 7/9/2026 NAME
K.Hemanth Kumar PROJECT
**OptiCrop – Smart
Agricultural
Production
Optimization Engine** MAX MARKS
— Project Title Page

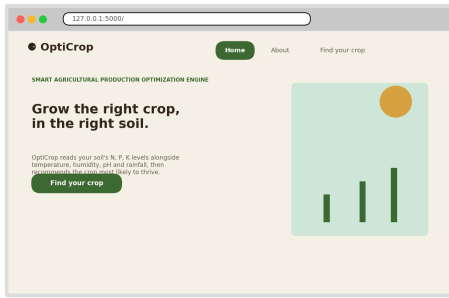
FIELD	VALUE
Project Title	OptiCrop – Smart Agricultural Production Optimization Engine
Name of the student	K.Hemanth Kumar
Roll no	232G1A3396
Name of the college	Anantha Lakshmi Institute of Technology & Sciences

 Introduction

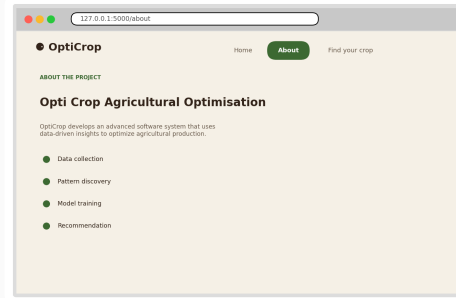
OptiCrop is an intelligent Machine Learning-based web application developed to recommend the most suitable crop for a given plot of land. The system helps farmers, agribusinesses, and agricultural researchers make faster, more accurate, and data-driven planting decisions by analyzing soil and climate readings and predicting the crop most likely to succeed.

The application leverages a curated agricultural dataset and multiple analytical techniques — including K-Means clustering to uncover soil/climate patterns and a Logistic Regression classifier for crop prediction — to identify the most suitable crop for a given input profile. The trained model is integrated into a Flask-based web application that lets users enter soil and climate readings and receive an instant crop recommendation.

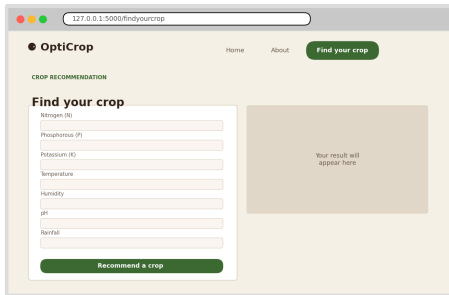
The system evaluates seven key factors before generating a recommendation: Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, soil pH, and rainfall.



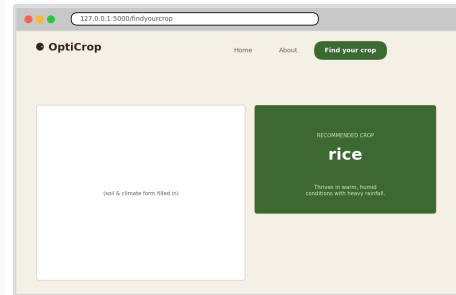
Home page — introduces OptiCrop and links to Find your crop



About page — explains the recommendation pipeline



Find your crop — soil & climate input form



Crop recommendation result page



Conclusion

OptiCrop demonstrates how Machine Learning can be applied to make crop selection faster, more consistent, and less dependent on guesswork. By analyzing soil nutrients and climate conditions — Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall — the system accurately recommends the crop best suited to a given plot, helping farmers plan with more confidence.

The integration of the trained Logistic Regression model with a Flask-based web application provides a simple, interactive platform for real-time crop recommendation. The system also groups agricultural conditions using K-Means clustering, offering additional insight into how different crops relate to one another by soil and climate profile.

OptiCrop serves as a scalable foundation that can be extended with additional ML models, larger and region-specific datasets, cloud deployment, and stronger data management as it grows.